INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME

CLASS

INDEX NUMBER

CHEMISTRY

8872/02

Section A: Structured Questions
Section B: Free Response Questions

18 September 2013

2 hours

Candidates answer Section A on the Question Paper
Additional Materials: Writing Papers
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
Write in dark blue or black pen.
You may use pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.
A Data Booklet is provided.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	**8**
2	**9**
3	**8**
4	**15**
Significant figures	
Handwriting	
Total	**40**

This document consists of **17** printed pages and **1** blank page.

For Examiner's Use

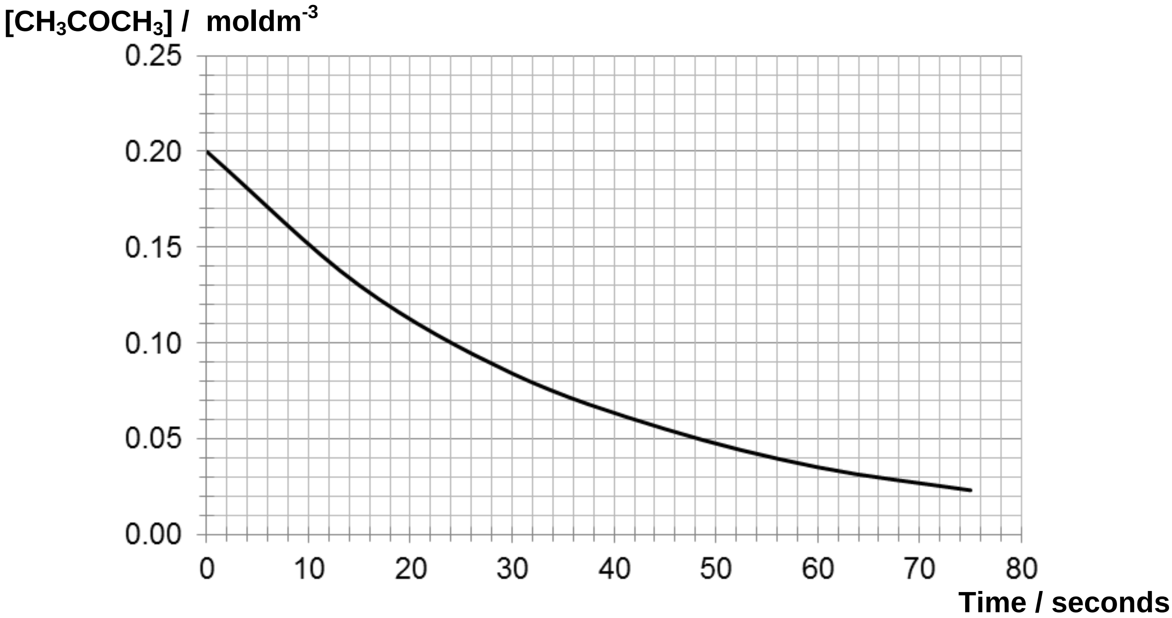
Section A

Answer ALL questions on the spaces provided.

1 The kinetics of the reaction between acetone and potassium cyanide was investigated.

An experiment was performed in which 0.200 mol dm^{-3} of acetone, CH_3COCH_3 was reacted with an excess of acid and sodium cyanide, NaCN. A graph of concentration of CH_3COCH_3 against time was plotted.

$$CH_3COCH_3 + H^+ + NaCN \longrightarrow CH_3C(OH)(CN)CH_3 + Na^+$$



(i) Using your graph, deduce the order of reaction with respect to acetone.

(ii) Why was the acid and sodium cyanide used in excess?

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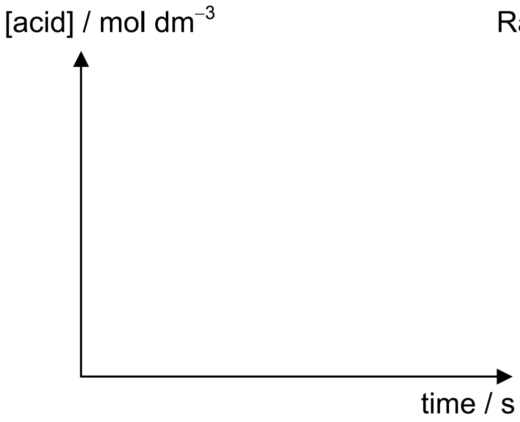
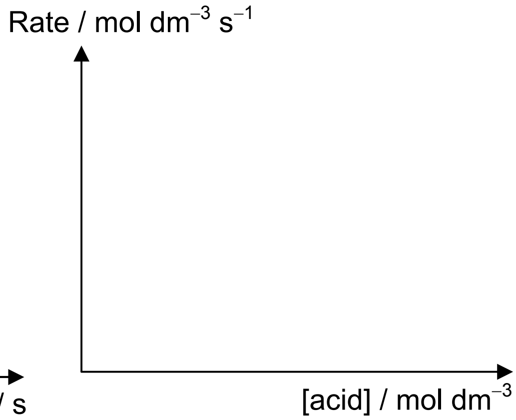
For Examiner's Use

(iii) The order of reaction with respect to the acid was zero.

Sketch, on the axes provided, the shape of the graphs relating to the information provided.



Graph 1



Graph 2

(iv) Given that the order of reaction with respect to sodium cyanide is one, write the rate law for this reaction.

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(v) Suggest with a reason whether pentan-2-one reacts more or less rapidly with NaCN as compared to acetone.

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[Total: 8]

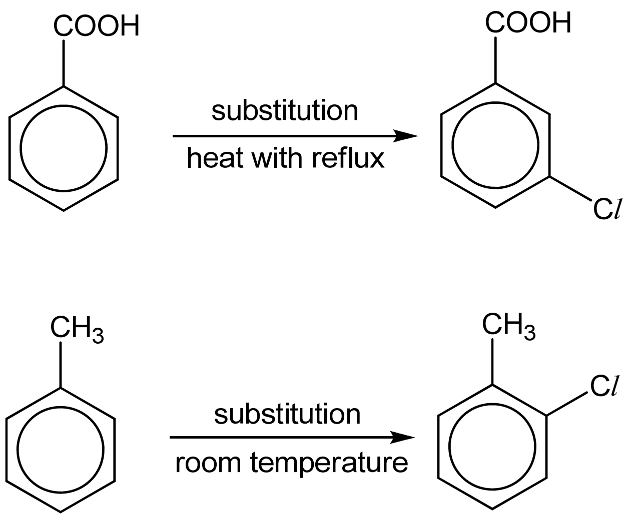
For Examiner's Use

2 Benzoic acid is a colourless crystalline solid that has antifungal abilities and is an important precursor for the synthesis of many organic compounds.

(a) Explain, with the aid of a diagram, why benzoic acid has an apparent molecular mass of 244 in an organic solvent such as benzene.

[2]

(b) Both benzoic acid and methylbenzene undergo aromatic substitution to produce the chloro-compounds shown:



(i) Suggest why a harsher condition is required for the substitution of benzoic acid.

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For Examiner's Use

(ii) How would you expect the acidity of 3-chlorobenzoic acid to compare with that of benzoic acid?

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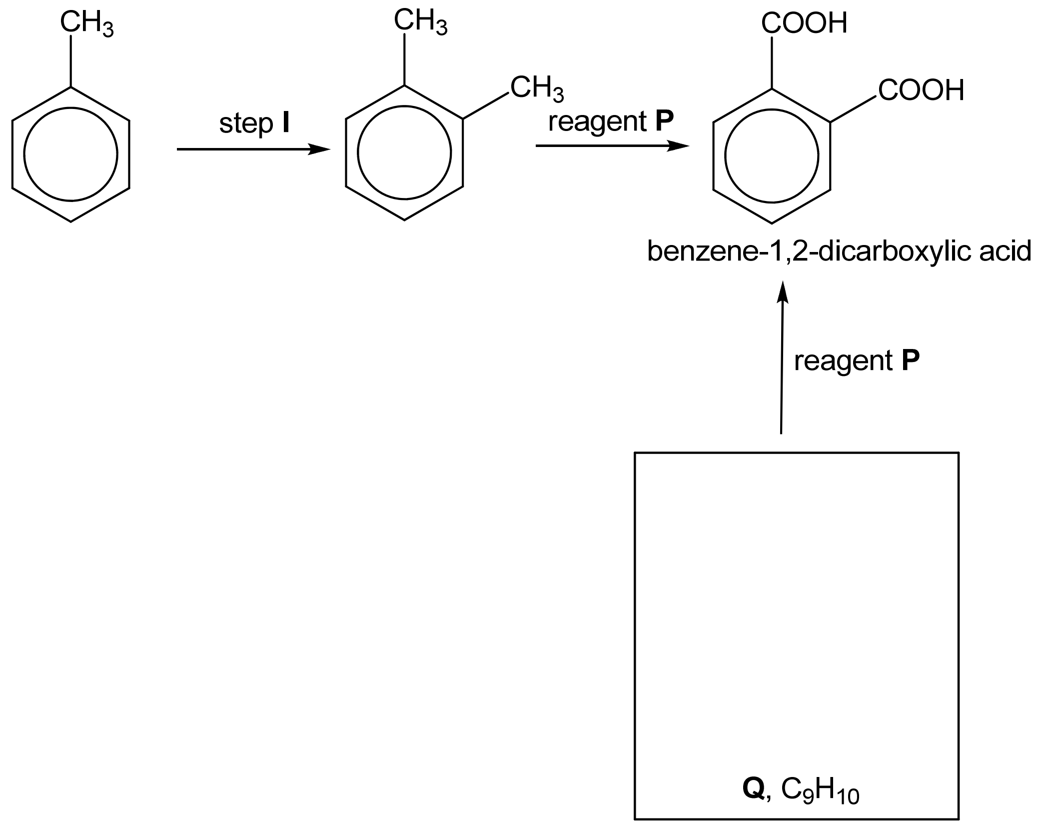
....[3]

(c) Benzoic acid is sparingly soluble in water. However, it dissolves readily to give a colourless solution upon the addition of a reagent, **N**. Identify reagent **N**.

....[1]

(d) Benzene-1,2-dicarboxylic acid can be produced via the following synthetic pathway.



(i) What type of reaction is step **I**?

....

(ii) Identify reagent **P**.

....

(iii) Suggest a structure for compound **Q** and draw its structural formula in the box provided.

[3]

[Total: 9]

For Examiner's Use

3 Phosphorus(V) chloride, PCl_5, is a white solid which sublimes at 160°C.

When gaseous phosphorus(V) chloride is heated in a closed container, the following equilibrium is established.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

(a) Sketch a graph to show how the rates of the forward and backward reactions change from the time phosphorus pentachloride is heated to the time the reaction reaches dynamic equilibrium. Label your two lines clearly.

[2]

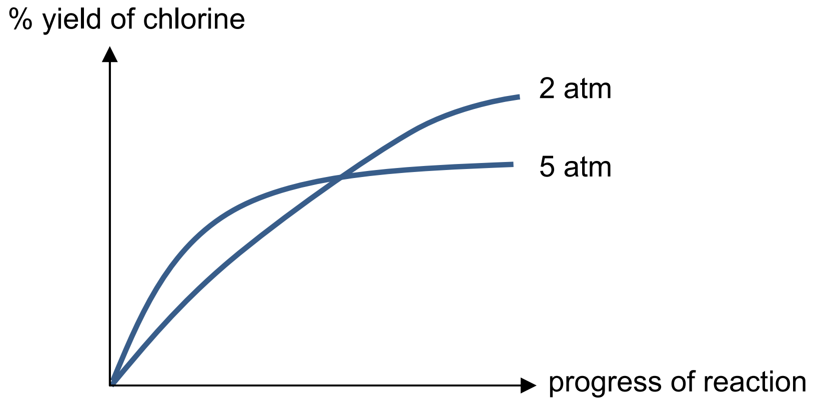
(b) In an experiment, 1.00 mol of PCl_5 vapour was heated in a closed 5.00 dm^3 flask at 500K until equilibrium had been established. The amount of chlorine collected was 0.508 mol.

Write an expression for the equilibrium constant and calculate its value.

[3]

For Examiner's Use

(c) The graph below shows how the percentage yield of chlorine from the above equilibrium varies as the reaction progresses at two different pressures.



(i) Account for the **differences** in the two graphs at different pressures.

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(ii) On the same axes, sketch a graph to show how the percentage yield of chlorine will vary at 5 atm in the presence of a catalyst.

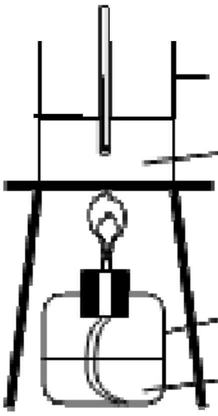
[3]

[Total: 8]

For Examiner's Use

4 **(a)** During the last Olympic Games, a torch relay around several countries took place before the torch was used to ignite the Olympic cauldron to mark the start of the Game. The fuel used to light the Olympic flame is propane. Propane is also used in blow torches, portable stoves and in outdoor heaters.

A propane portable stove was used during a barbecue to boil water for making of soup. It took 10 minutes to bring the water to boil.



The following information was known.

Mass of metal container / g	850
Mass of metal container containing water / g	1600
Initial temperature of water / °C	32
Final temperature of water / °C	100
Specific heat capacity of water / J g^{-1} K^{-1}	4.18

Besides the above information, it was also known that the theoretical standard enthalpy change of combustion of propane is −2220 kJ mol^{-1}.

(i) Define the *standard enthalpy change of combustion of propane*.

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....

(ii) Calculate the mass of propane needed to raise the temperature of the water during the barbecue.

For Examiner's Use

Another value of the standard enthalpy change of combustion of propane, ΔH_c, can be calculated using bond energies and the energy cycle below.

$$\begin{tikzcd}
C_3H_8(g) + 5O_2(g) \arrow[rr, "\Delta H_c"] \arrow[dr, "+6460\ \text{kJ mol}^{-1}"'] & & 3CO_2(g) + 4H_2O(g) \arrow[dl, "\Delta H_1"] \\
& 3\ C(g) + 8\ H(g) + 10\ O(g) &
\end{tikzcd}$$

(iii) Use suitable bond energies given in the *Data Booklet* to calculate ΔH_1.

(iv) Hence, calculate ΔH_c.

(v) Suggest why the value of ΔH_c you calculated differs from the theoretical value.

....

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....[7]

For Examiner's Use

(b) Most outdoor heaters are powered by the combustion of propane gas stored in small cylinders. At room temperature and pressure, a typical outdoor heater designed to produce 15 kW of energy runs from a cylinder containing 13 kg of propane.

As pure propane gas is odourless, small amounts of another compound are usually added so that gas leaks from propane cylinders can be detected. An example of such an odorant is ethanethiol, C_2H_5SH which has an odour that resembles that of onion. Ethanethiol is chosen since the human nose can detect its presence at levels of only about 0.02 moles of it per billion (10^9) moles of propane.

(i) Calculate the total amount of heat energy released by the complete combustion of all the propane in a cylinder used in an outdoor heater.

(ii) Using your answer to **(b)(i)**, calculate the time needed to empty one cylinder of propane to produce 15 kW of energy. You are given that 15kW is equivalent to 15 kJ s^{-1}.

(iii) Hence, calculate the rate, in $cm^3\ s^{-1}$, at which propane must leave the cylinder in order to produce 15 kW of energy.

For Examiner's Use

(iv) Calculate the volume of carbon dioxide evolved when 13 kg of propane was completely burnt in an outdoor heater.

(v) Calculate the minimum mass of ethanethiol which must be added to 13 kg of propane so that it can be detected by the human nose in the event of a propane gas leak.

[5]

(c) Besides being used as fuel, propane can also be used to synthesise other organic compounds.

Outline the synthesis of propan-1-ol from propane. State clearly the reagents and conditions used and draw any intermediate product(s) formed.

[3]

[Total: 15]

Section B

Answer **two** of the three questions in this section on separate answer paper.

5 **(a)** Ethanal is one of the most important aldehydes occurring widely in nature and is also the cause of hangovers from alcohol consumption. It is produced on a large scale industrially via the Wacker Process as shown.

$$2\ CH_2{=}CH_2 + O_2 \longrightarrow 2\ CH_3CHO$$

(i) State the hybridisation of each carbon atom in ethene and draw a diagram to show the hybridised orbitals around one of these carbon atoms.

(ii) Describe, in terms of orbital overlap, the bonding of the two carbon atoms of the C=C bond in ethene.

[4]

(b) The Wacker reaction is accelerated by palladium catalysts.

(i) Define *catalyst*.

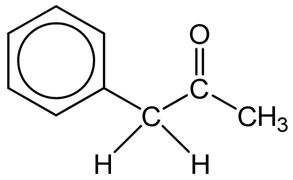
(ii) Explain with the aid of a diagram that shows the distribution of molecular energies, how the addition of a catalyst accelerates the reaction.

(iii) In a college laboratory, ethanal can be synthesised from ethanol. Briefly outline how you would carry out this reaction.

(iv) Explain why ethanal can be prepared in the way you described in **(b)(iii)**, in terms of the volatility of ethanal and ethanol.

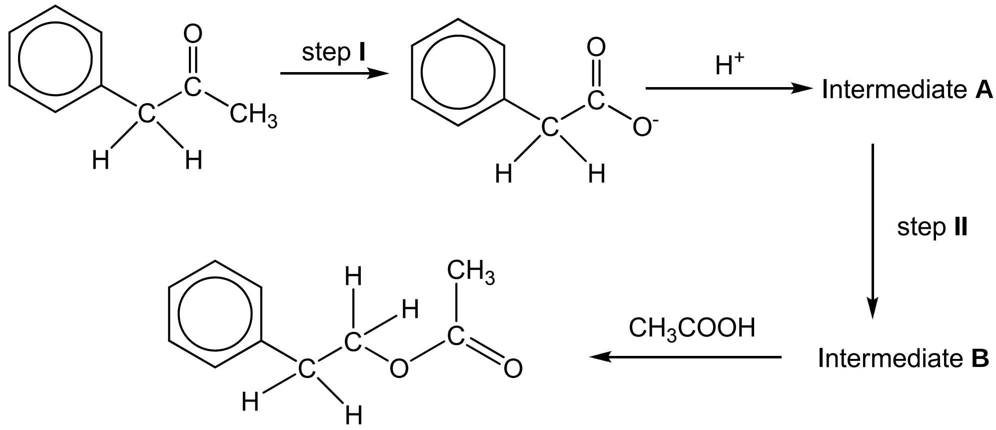
[8]

(c) **(i)** Another common carbonyl compound used in the industry is phenylacetone, which is an intermediate used to produce pesticides and anticoagulants.



phenylacetone

Compound **C** can be synthesised from phenylacetone.



Compound **C**

Complete the synthetic route by giving the structures of intermediates **A** and **B** as well as the reagents and conditions for steps **I** and **II**.

(ii) Suggest the structures of the product and type of reaction undergone when phenylacetone reacts with the following reagents.

- HCN, trace OH^-
- hydrazine, $H_2N\text{-}NH_2$

[8]

[Total: 20]

6 **(a)** The elements of the third period of the Periodic Table, sodium to sulfur, all form chlorides by direct combination.

(i) When dry chlorine is passed over heated aluminium powder in a long hard-glass tube, a vapour is produced which condenses to a solid in the cooler parts of the tube. At low temperatures, the vapour has a M_r of 267.

State one observation you would make during this reaction.

(ii) Suggest the molecular formula of the vapour, and draw a dot-and-cross diagram to illustrate its bonding.

(iii) The solid reacts with water in two separate ways.

- When a few drops of water are added to the solid, steamy white fumes are evolved and a white solid remains, which is insoluble in water.
- When a large amount of water is added to the solid, a clear, weakly acidic solution results.

Write equations, including state symbols, for these two reactions.

[7]

(b) Lattice energy can be used as a measure of the energetic stability of ionic compounds.

(i) Define the *lattice energy of silver fluoride*.

The table below shows numerical values of lattice energies for silver fluoride and silver iodide. These have been determined from experimental data or theoretically calculated.

Compound	Experimental value / kJ mol^{-1}	Theoretical value / kJ mol^{-1}
AgF	–967	–953
AgI	–889	–808

(ii) By quoting appropriate data from the Data Booklet, explain why the lattice energy of silver iodide is less exothermic than silver fluoride.

(iii) Silver fluoride and silver iodide have the same crystal structure. There is close agreement between the experimental and theoretical values of lattice energy for AgF but not for AgI. Suggest a reason for this.

[4]

(c) Compound **C** has the empirical formula CH_2O and M_r of 90. There is no reaction when **C** is treated with $NaHCO_3$. When 0.600 g of **C** is reacted with an excess of Na, 160 cm^3 of H_2, measured at room temperature and pressure, is produced. Treatment of **C** with 2,4-dinitrophenylhydrazine reagent produces an orange solid. When **C** is warmed with Fehling's reagent, a brick red precipitate is formed.

The structure of **C** has the following feature:

- No carbon atom has more than one oxygen atom joined to it.

Suggest a structure for **C**, explaining your reasoning.

[7]

(d) Compound **C** can be both oxidised and reduced.

(i) Give the structural formula of the compound formed when **C** is reacted with $NaBH_4$ under suitable conditions.

(ii) Give the structural formula of the compound formed when **C** is heated under reflux with acidified $K_2Cr_2O_7$.

[2]

[Total: 20]

7 **(a)** When sodium is burned in air, a mixture of sodium oxide, Na_2O, and sodium peroxide Na_2O_2, is formed. The mixture reacts with water according to the following equations.

$$Na_2O + H_2O \longrightarrow 2NaOH$$
$$Na_2O_2 + 2H_2O \longrightarrow 2NaOH + H_2O_2$$

The following information will allow you to calculate the relative amounts of the two oxides produced when sodium is burned.

- The mixture obtained by burning a sample of sodium was dissolved in distilled water and made up to 100 cm^3 to give solution **D**.
- A 25.0 cm^3 portion of solution **D** was titrated with 0.100 mol dm^{-3} HC*l*. 22.5 cm^3 of acid was required to reach the end-point.
- The H_2O_2 content of solution **D** was found by titration of another 25.0 cm^3 portion with 0.0200 mol dm^{-3} $KMnO_4$.

The following reaction occurs.

$$2MnO_4^- + 5H_2O_2 + 6H^+ \longrightarrow 2Mn^{2+} + 5O_2\ 8H_2O$$

10.0 cm^3 of $KMnO_4$ solution was required to reach the end-point.

(i) Using the results of the HC*l* titration calculate the total amount in moles of NaOH in **100 cm^3** of solution **D**.

(ii) Using the results of the $KMnO_4$ titration, calculate the amount in moles of H_2O_2 in **100 cm^3** of solution **D**.

(iii) Hence calculate the amount in moles of Na_2O_2 formed during the burning of the sodium sample.

(iv) Using your result in **(a)(iii)**, calculate the amount in moles of Na_2O formed during the burning of the sodium sample.

[6]

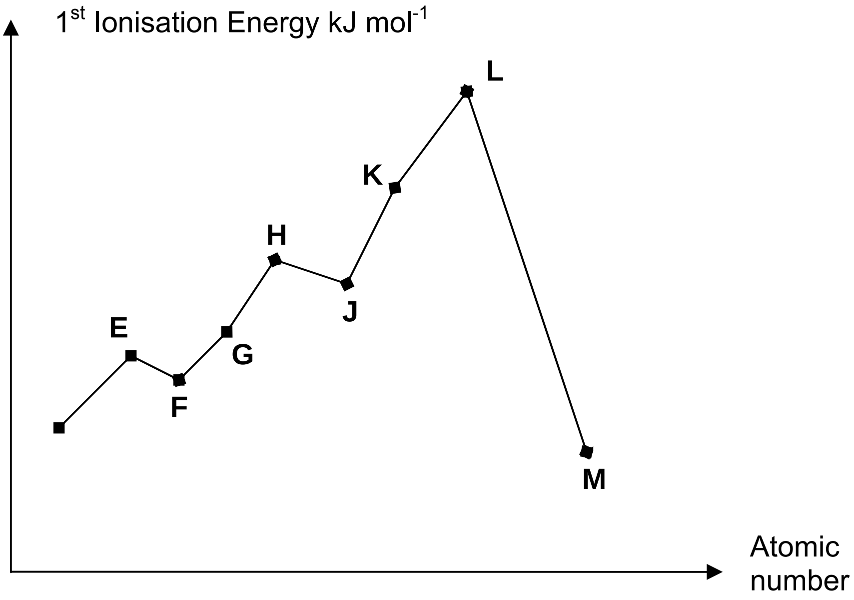
(b) A buffer solution can be prepared by mixing 500 cm^3 of 4 mol dm^{-3} of CH_3COOH in 500 cm^3 of 0.5 mol dm^{-3} of NaOH.

(i) Calculate the pH of NaOH before mixing is carried out.

(ii) By using suitable equations, describe how the above buffer solution works.

[4]

(c) The graph below shows the first ionisation energy of eight elements with consecutive proton number.



(i) State the valence electronic configuration of element **F**.

(ii) Give the formula of the oxide that **F** is likely to form.

(iii) Hence or otherwise, explain the drop in first ionisation energy from **E** to **F**.

[4]

(d) **(i)** Sketch a graph of the **melting point of oxides** of elements from sodium to sulfur.

(ii) Explain as fully as you can why the melting point varies in the way shown.

[6]

[Total: 20]

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